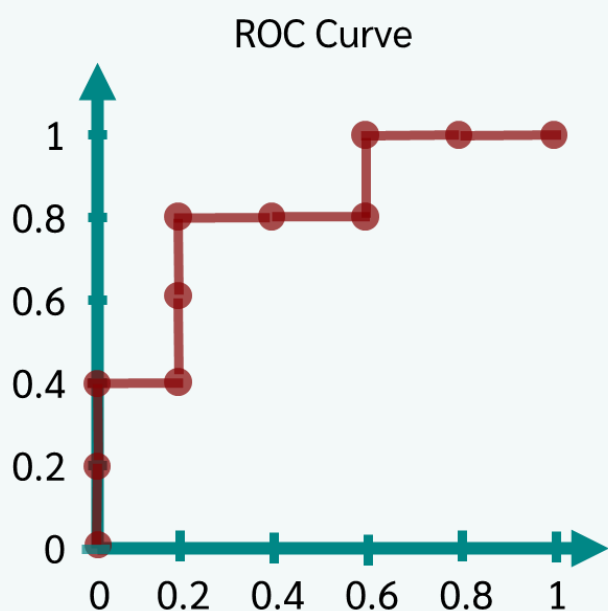


ROC CURVE

Playbook

Theory & Example

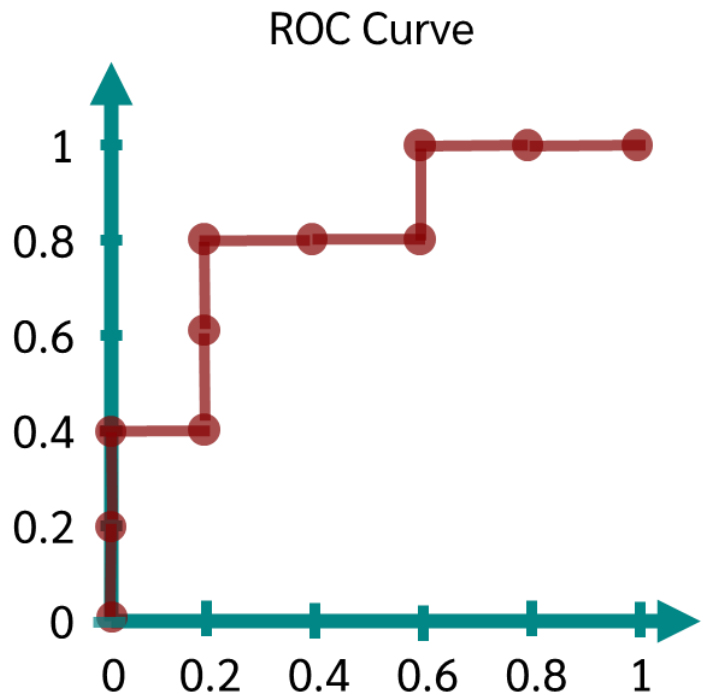




What is a **ROC Curve**?

ROC stands for **R**eceiver **O**perating **C**haracteristic.

A **ROC Curve** is a **graphical representation** of the performance of a **binary classification model** for all **classification thresholds**.



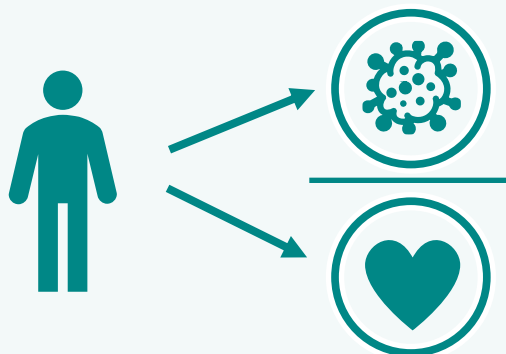


Example:

We would like to **classify**
based on a **screening**,



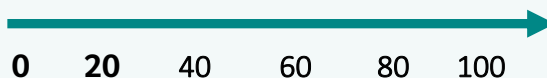
whether a person **has cancer**
or **not**.



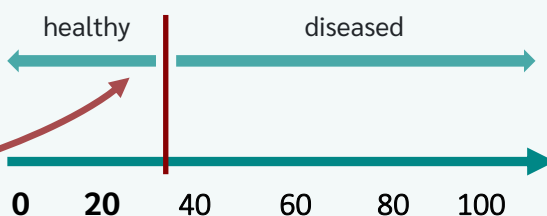
This classification is done
with the help of a certain
blood value,



where **high values** indicate
cancer.

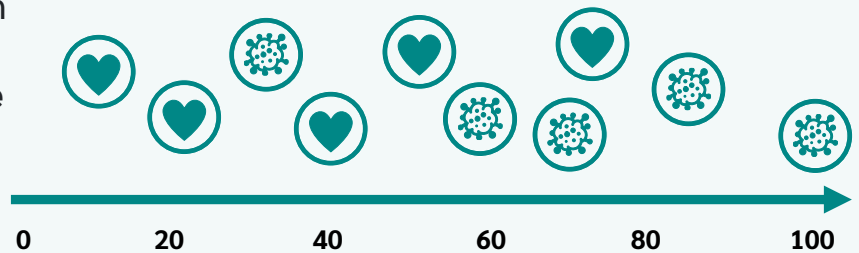


The question now is
what value we choose
as the **classification**
threshold.

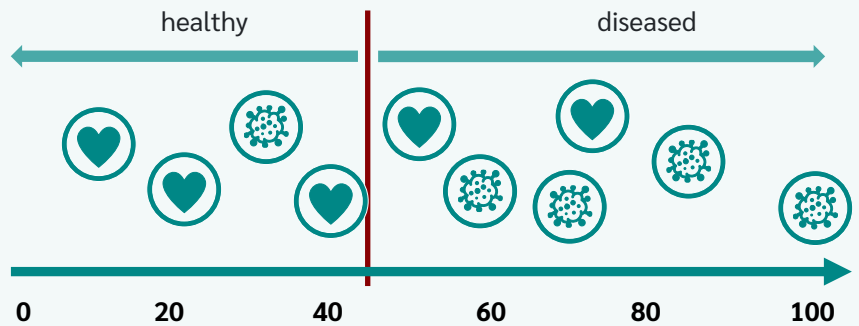


So at what level do we
predict a disease?

For this, we get **data from 10 people** about how high the **blood level** is and whether there is a **disease or not**.

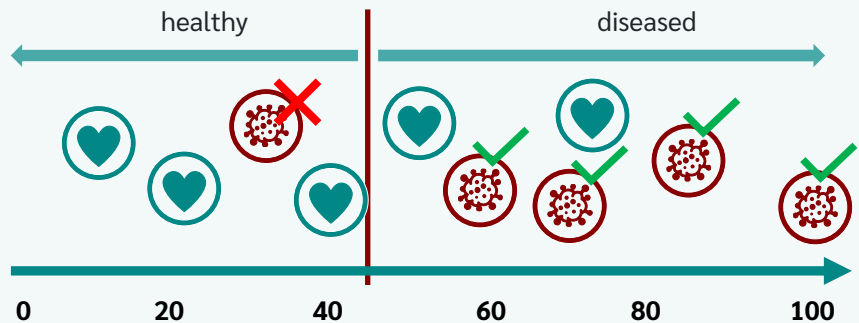


We could now choose a **classification threshold** of **45**, for example.



In this case, of the **5 diseased individuals**,

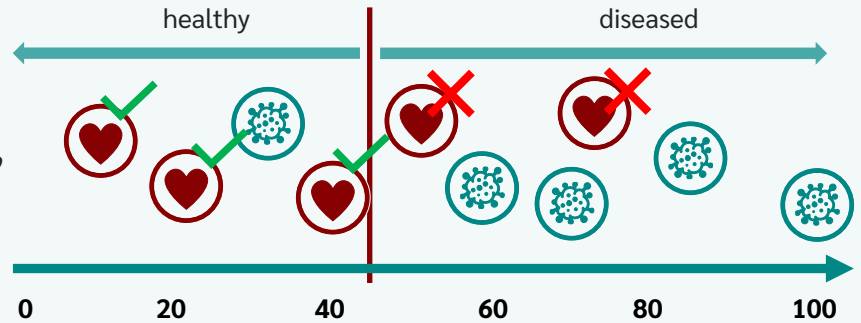
we would **correctly** classify **4** as "diseased" and **1** **incorrectly** as "healthy".



➡ So, we correctly classified **4 out of 5** as "diseased". This value is called the **True Positive Rate (TPR)** and is equal to the **sensitivity**.

On the other hand,
of the **5 healthy**
individuals, we mis-
classified 2 as "diseased"

and **correctly** classified
3 as "healthy".



So, we misclassified **2 out of 5** as "diseased".
This value is called the **False Positive Rate (FPR)**
and is $1 - \text{specificity}$.

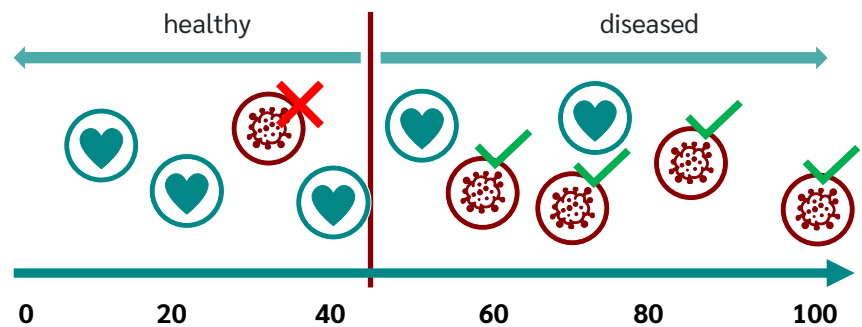
So, for a **threshold of 45**, we get a
True Positive Rate of $4/5$ i.e. **0.8**
and a **False Positive Rate** of $2/5$ i.e. **0.4**



How do we calculate TPR and FPR?

$$\text{TPR (True Positive Rate)} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

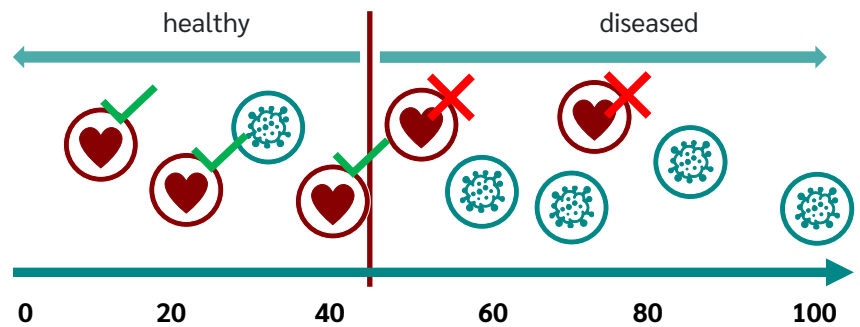
* The true positives are the persons correctly classified as diseased and the false negatives are the persons incorrectly classified as healthy.



$$\text{TPR (True Positive Rate)} = \frac{4}{4 + 1} = \frac{4}{5} = 0.8$$

$$\text{FPR (False Positive Rate)} = \frac{\text{False Positives}}{\text{False Positives} + \text{True Negatives}}$$

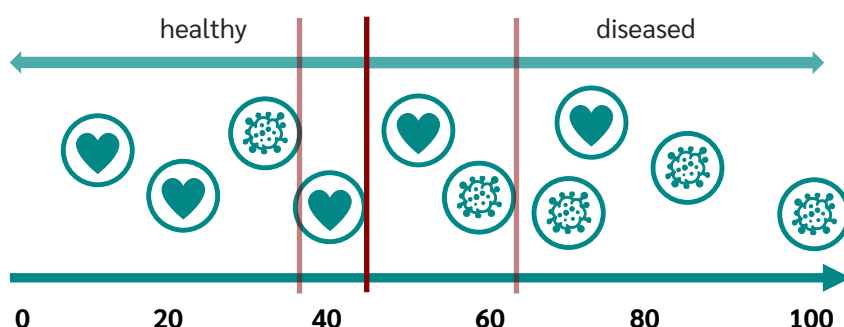
* The false positives are the healthy individuals misclassified as diseased and the true negatives are the individuals correctly classified as healthy.



$$\text{FPR (False Positive Rate)} = \frac{2}{2 + 3} = \frac{2}{5} = 0.4$$

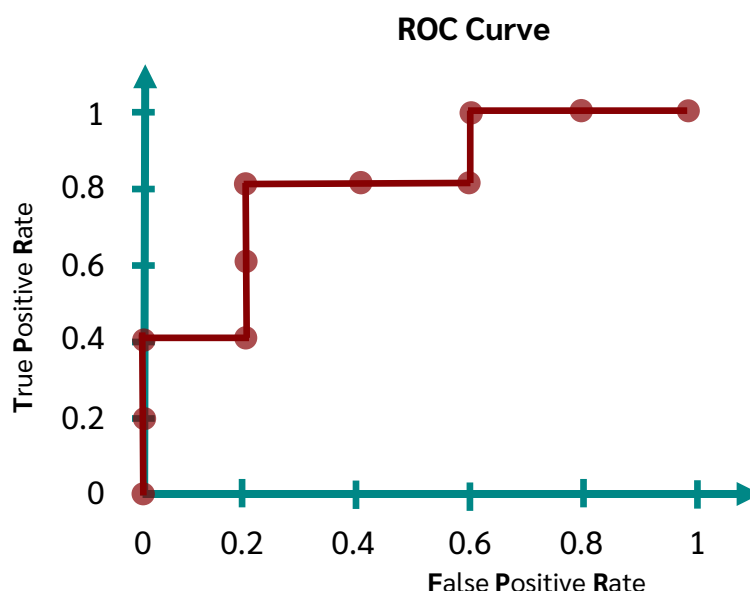


How do we calculate TPR and FPR for each threshold?



It is precisely these **two values** that are then plotted on the **ROC Curve**.

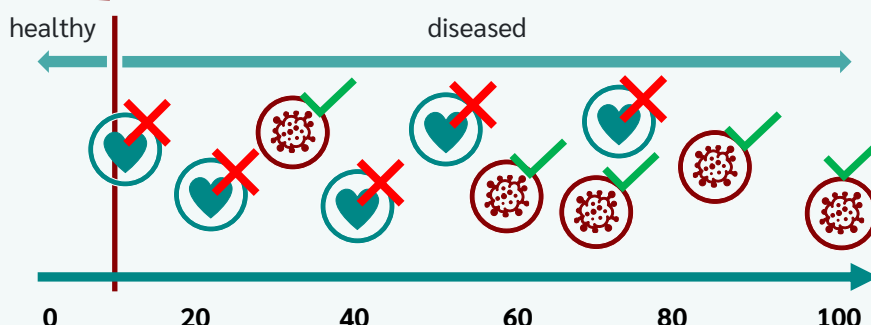
The **True Positive Rate** is plotted on the **y-axis** and the **False Positive Rate** on the **x-axis**.



Example:

Now let's draw the complete ROC Curve for our example!

If we choose the **threshold value** to be very small, i.e., pushed all the way to the **left**,



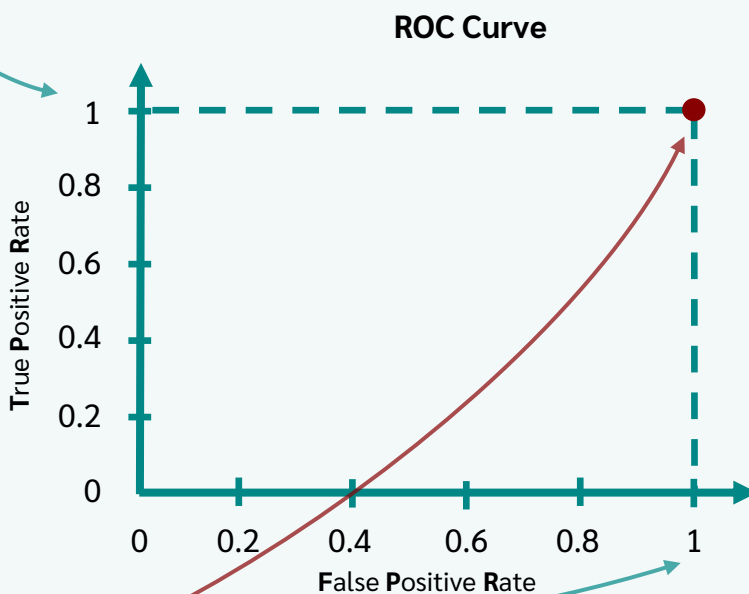
we **correctly classify** all **5 diseased** individuals.

Our **True Positive Rate** is thus **5 out of 5**, i.e. **1**.

In the same way, however, we also **misclassify** all **5 healthy** people as "diseased".

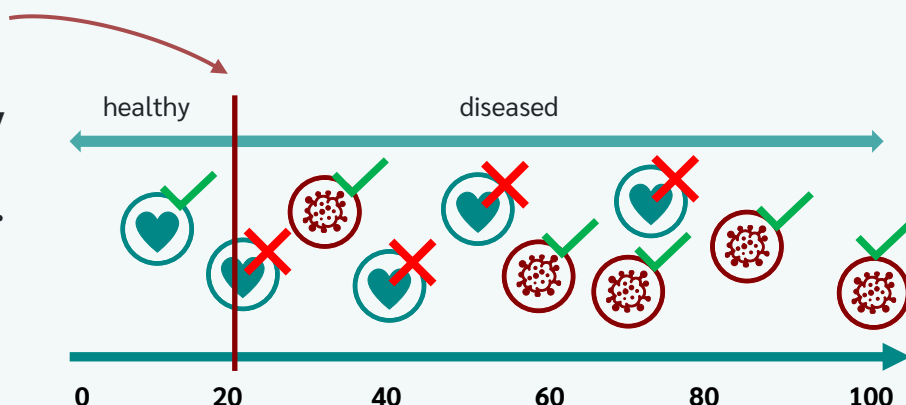
Our **False Positive Rate** is thus **5 out of 5**, i.e. **1**.

With that, we have the **first point**.

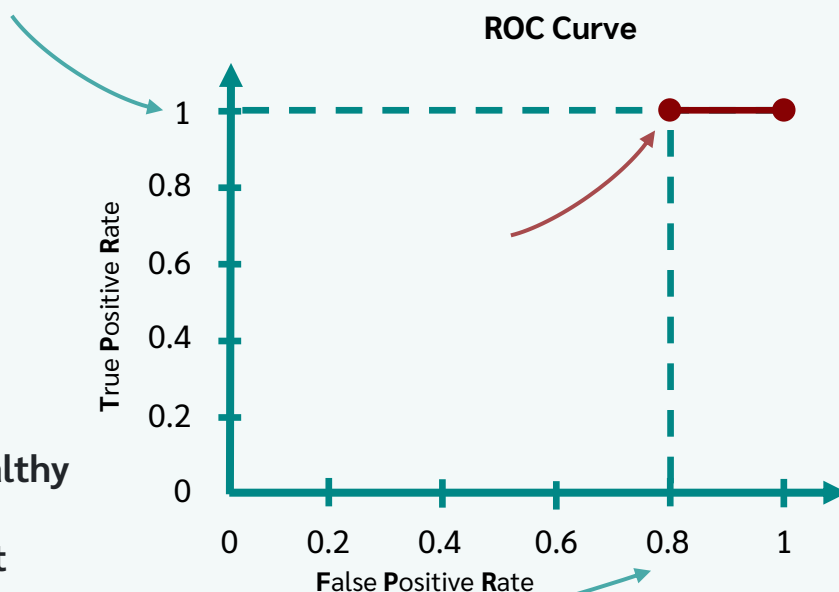


Now we can push the **threshold** further.

Here we still **correctly** classify all **5** diseased people as “diseased”.



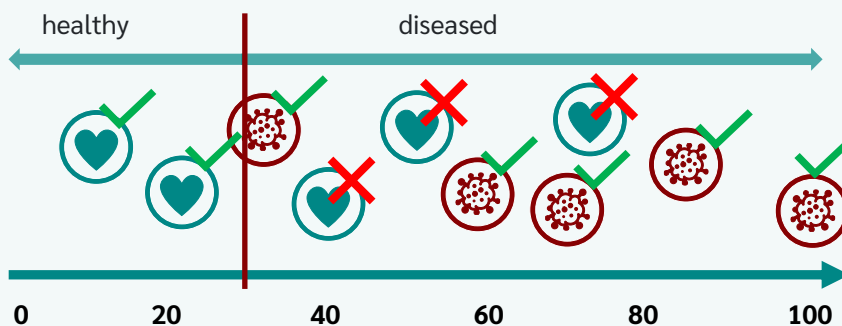
So, we continue to have a **True Positive Rate 5/5**.



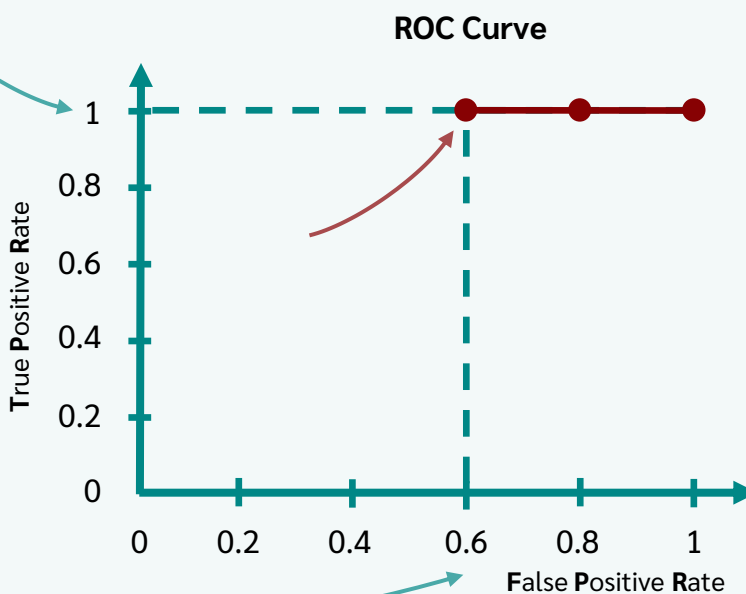
However, of the **5** healthy individuals, we now **misclassify** only **4** out of **5** as “diseased.”

So, we have 4 out of 5 or **0.8**.

At the next **threshold**,
we still have a **True
Positive Rate of 1.**

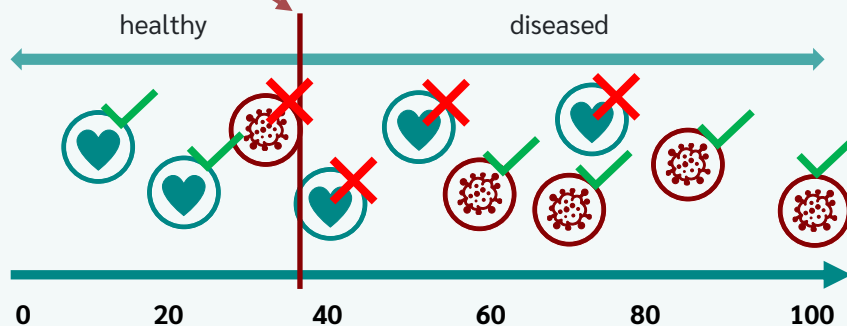


All **5 diseased** people are
correctly classified.

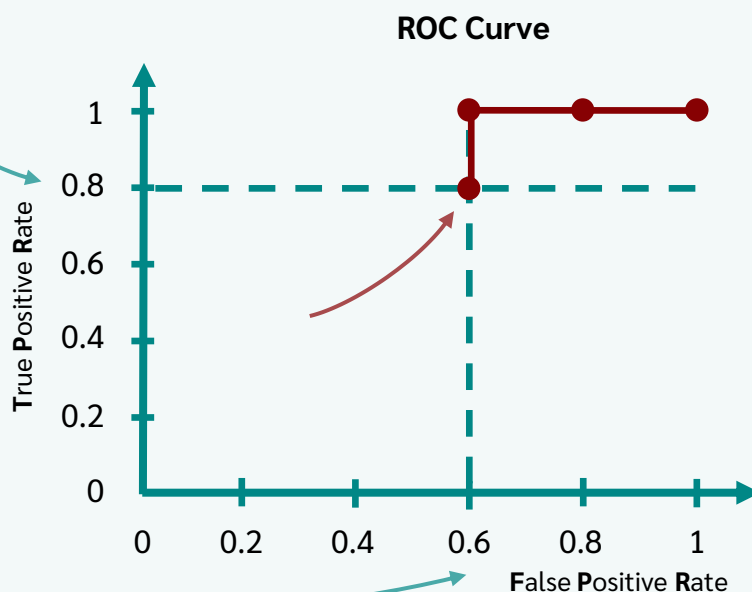


And a **False Positive Rate
of 3/5, i.e. 0.6**

At the next **threshold**,
a diseased person is **mis-**
classified as "healthy"
for the first time.

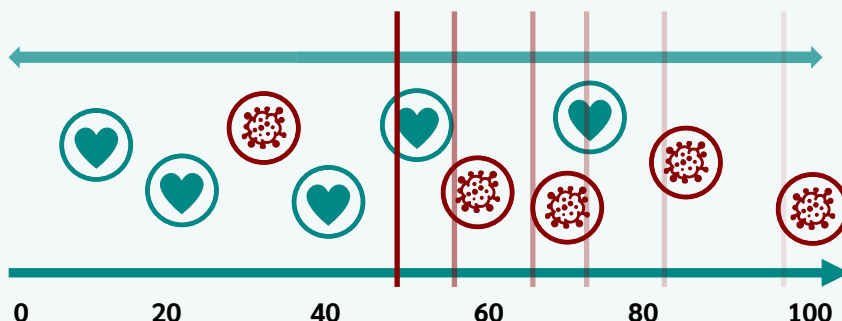


We therefore obtain a
True Positive Rate
of $4/5$, i.e. **0.8**



And a **False Positive Rate**
of $3/5$, i.e. **0.6**

We can now do that
for all other thresholds.

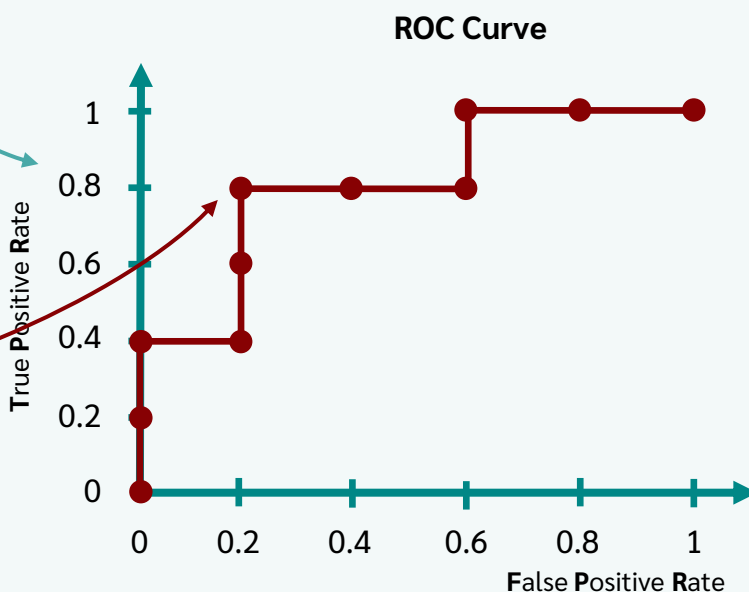


And get our **finished**
ROC Curve.

At this point,
for example

80% of the diseased
individuals were correctly
classified as “diseased”

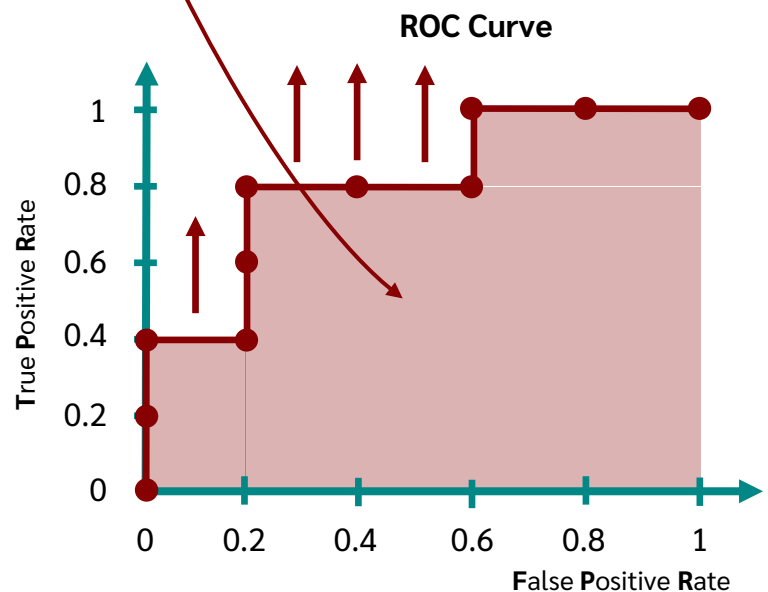
and **20%** of the healthy
individuals were incorrectly
assigned “diseased”.



Using the ROC Curve, we can **compare different classification methods**.



A **classification model** is better the higher the curve. Therefore, the **larger the area under the curve**, the **better** the classifier is.



And it is precisely this area that is **reflected by the AUC value**, the Area under the Curve.

The **AUC value** varies between **0 and 1**.

The **larger the value**, the **better** the classifier.

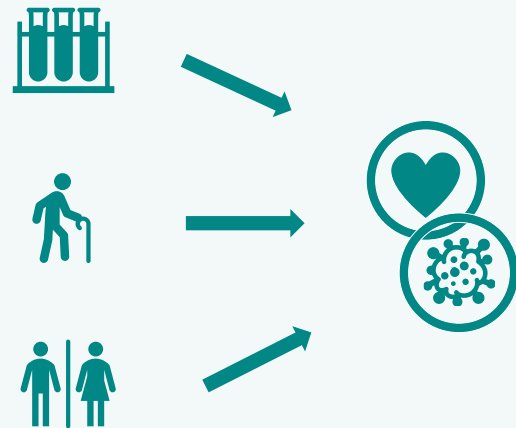
What about the **ROC Curve** and **logistic regression**?



For example:

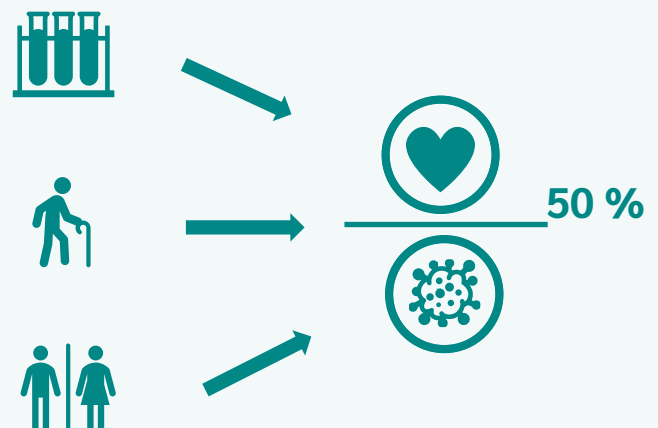
We could build a new **classification model** using a **logistic regression**.

Here we could use, in addition to the **blood value**, **age** and **gender** of each person.



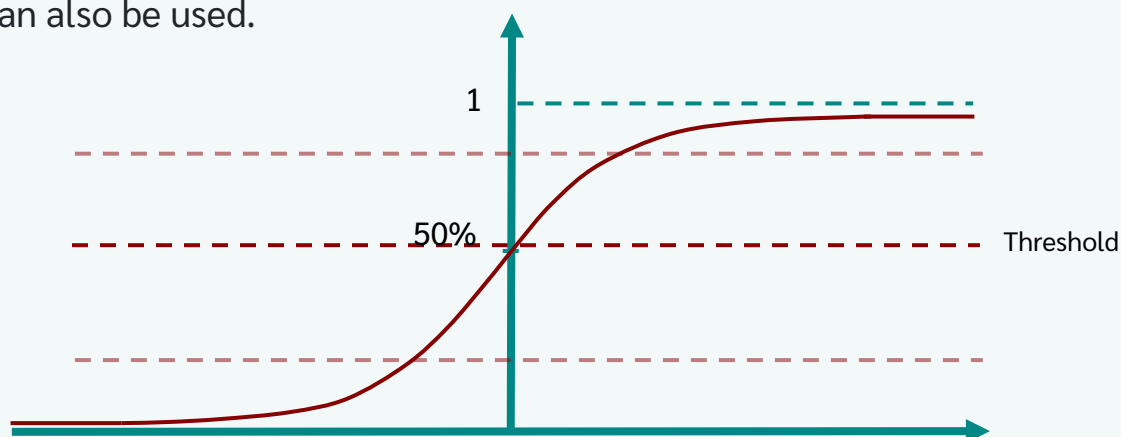
 In a **logistic regression**, the estimated value is then **how likely** it is that a particular **person is diseased**.

Very often, **50%** is then simply taken as the **threshold to classify** whether a person is diseased or not.

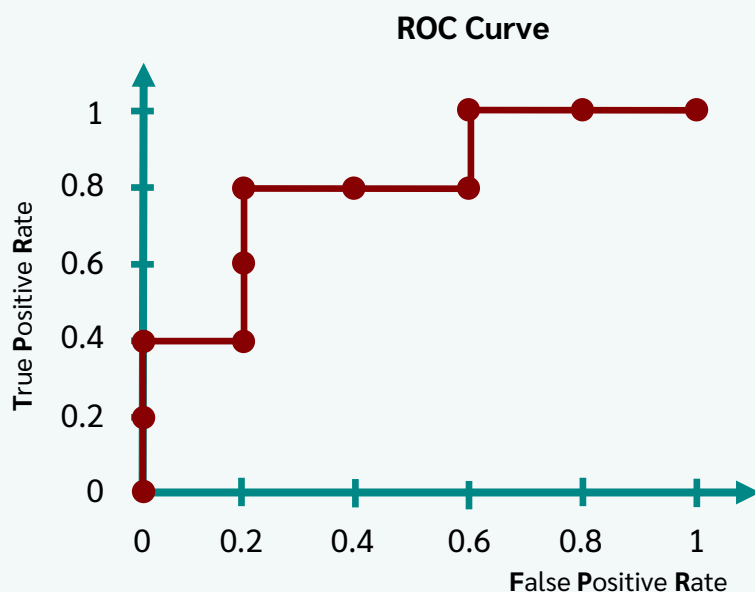




But of course, this does not have to be!
Any other threshold can also be used.



Therefore, even with **logistic regression**, we can construct a **ROC Curve** for the different **thresholds**.





How do we get a **ROC Curve** with **DATAtab**?

Simply copy our data into this [table](#) and click on **ROC**.

DATAtab [Statistics Calculator](#) [Survey](#) [Tutorials](#) [Book](#) [Pricing](#) Login DE Q

Online Statistics Calculator 100% data security ©

Clear Table Export / Import Transform data Settings

	metric	nominal
Cases	Blood value	Diseased
1	35	yes
2	10	no
3	18	no
4	40	no
5	55	no
6	58	yes
7	72	yes
8	88	no
9	92	yes
10	95	yes
11	23	no
12	54	yes
13	54	yes
14	23	no
15	22	no

Descriptive Charts Hypothesis tests Correlation Regression Mediation/Moderation PCA Reliability Cluster Decision Tree **ROC**

Survival Analysis Equivalence & Non-inferiority Association Rules Process Capability Measurement Systems Analysis

State Variable: ☒ Diseased Test Variable: ☒ Blood value

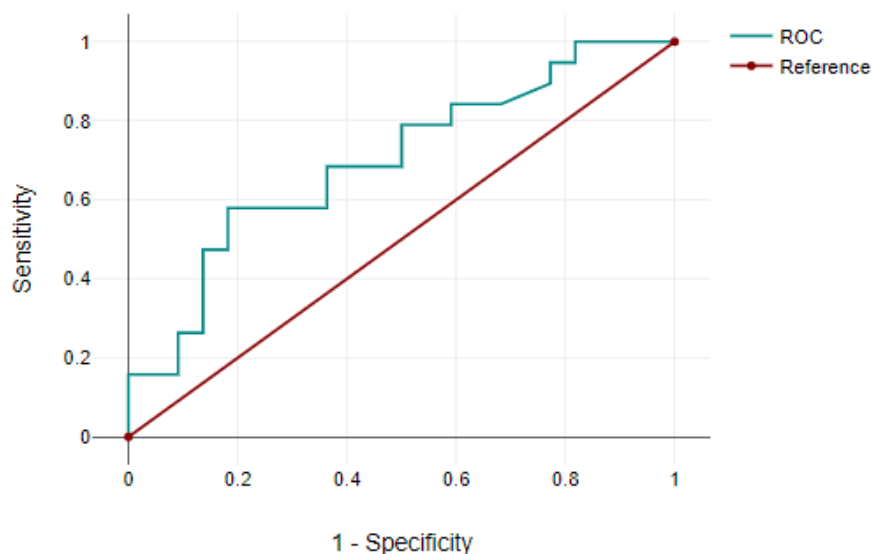
[GO TO DATAtab](#)

Receiver Operating Characteristic

ROC-Curve

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ROC Curve (AUC: 0.699)



More about

ROC CURVE

on our website datatab.net

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